

```

clc; clear; close all;

%% --- Input Parameters ---
fp = 1000;      % High-pass cutoff frequency (Hz)
fs = 8000;      % Sampling frequency (Hz)
N = 50;         % Filter order
wc = fp/(fs/2); % Normalized cutoff (0-1)

%% --- FIR High Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'high', rectwin(N+1));
disp('--- High-Pass Coefficients (Rectangular Window) ---');

```

```

--- High-Pass Coefficients (Rectangular Window) ---

```

```

disp(b_rect');

```

```

-0.0089
  0
  0.0097
  0.0143
  0.0106
  0
-0.0117
-0.0175
-0.0131
  0
  0.0148
  0.0225
  0.0171
  0
-0.0202
-0.0314
-0.0247
  0
  0.0318
  0.0524
  0.0445
  0
-0.0741
-0.1572
-0.2223
  0.7407
-0.2223
-0.1572
-0.0741
  0
  0.0445
  0.0524
  0.0318
  0
-0.0247
-0.0314
-0.0202
  0
  0.0171
  0.0225
  0.0148
  0
-0.0131
-0.0175

```

```
-0.0117  
0  
0.0106  
0.0143  
0.0097  
0  
-0.0089
```

```
%% --- Frequency Response Plot ---  
figure;  
freqz(b_rect,1);  
title('High-pass FIR using Rectangular Window');
```

